

DIANN converter program:

Workflow:

The DIANN output file contains all the information per annotated peptide per file in a separate row.

To use SafeQuant, we must have each annotated peptide and all the corresponding values for each file in one row. DIANN-SafeQuant-report_convert_protein_20230807.R was created to convert the DIA-NN "report.tsv" file to a format that can be used for SafeQuant and is not part of SafeQuant but an individual tool.

DIANN-SafeQuant-report_convert_protein_20230807.R will convert the "report.tsv" according to protein annotations. All the annotated peptides assigned to the same protein will be presented in the same row for all the different measurements. Each table entry will contain information about the protein, and the corresponding information from each measurement will be described in individual columns. The output contains information about the total number of precursors measured, the number of razor precursors, and the quantity from each measurement of the corresponding protein.

Inside the converter, the data are filtered out according to the Q value. By default, the "Global.PG.Q.Value" is used where anything higher than 0.01 will be excluded from the table. There is also the possibility to filter out the data using the "Protein.Q.Value".

The data will be reorganized according to the "Protein.Group" column, and the needed information will be gathered for each file. Razor precursors are the precursors' assigned peptide sequence that could fit more than one protein group. The RAZOR'S OCCAM approach was applied to all the razor precursors, and they set to one protein from all the possible protein groups, which had the most unique or abundant peptides. There are different types of categories:

- 1) Only one of the annotated proteins has a unique peptide. For example, the Peptide sequence "TAEALRGELVAAR" was assigned to two different groups, "Q7WXL9" and "Q0KAN7". In the report, unique peptides were found only for the "Q7WXL9"; as a result, all the razor precursors between the "Q7WXL9" and "Q0KAN7" were re-assign to the "Q7WXL9".
- 2) All the annotated proteins have unique peptides.
 - a. Annotation to the protein with the most peptides. For example, the Peptide sequence "VCGHYTQVVWR" was assigned to two different protein groups, "Q1CWK2" and "Q1CWK3". In the report, the "Q1CWK2" has only two unique peptides, and the "Q1CWK3" has 10; for this reason, all the razor precursors between the two proteins were assigned as peptides for the "Q1CWK3".
 - b. An equal number of unique peptides for all the proposed annotated proteins for a razor precursor. For example, the Peptide sequence "AALEWVPASAHPMVMR" was assigned to two different protein groups, "Q0KAG4" and "Q0K8F3". Both of the annotated proteins have, in total, 8 unique annotated peptides within the report. In this case, the razor precursors were assigned to the protein with the maximum sum of the area from the unique peptides, which is "Q0KAG4" with a maximum area of 93787357 over 38246265 "Q0K8F3".

- 3) Neither annotated protein has unique peptides. If no unique peptide is found for all the proposed proteins in a razor precursor, it will not be assigned to one but will stay with all of them.

The final output of the converter is named “report_SQ_final_convert.tsv” and can be used by SafeQuant as an input.

Application:

To run DIANN-SafeQuant-report_convert_protein_20230807.R script, use the following commands in the command line:

```
H:\>C:\Users\Angelidou\Documents\RA\4.1.2\bin\Rscript.exe D:\proteomics\Github\SafeQuant_p\DIANN_converter\DIANN-SafeQuant-report_convert_protein_20230807.R -h
Usage: DIANN-SafeQuant-report_convert_protein [options]

Options:
  -i INPUTFILE, --inputFile=INPUTFILE
                        I/O: Input file: DIANN file report.tsv
  -l RESULTSFILELABEL, --resultsFileLabel=RESULTSFILELABEL
                        I/O: results file name [default report_SQ_final_convert.tsv]
  --QValuefilter= GLOBAL: GLOBAL.PG.Q.VALUE
                        PROTEIN: PROTEIN.Q.VALUE
                        FILTER: --QV 'Global', 'Protein' [default Global]
  -h, --help
                        Show this help message and exit

Examples:
>Rscript DIANN-SafeQuant-report_convert_protein_20230807.R -i /path/to/report.tsv
```

Other examples:

```
> Rscript "c:\Program Files\DIANN_converter\DIANN-SafeQuant-report_convert_protein_20230807.R" -i "c:\Program Files\DIANN Run\Project Name\report.tsv"
```

```
> Rscript "c:\Program Files\DIANN_converter\DIANN-SafeQuant-report_convert_protein_20230807.R" -i "c:\Program Files\DIANN Run\Project Name\report.tsv" -QV Protein
```

```
> Rscript "c:\Program Files\DIANN_converter\DIANN-SafeQuant-report_convert_protein_20230807.R" -i "c:\Program Files\DIANN Run\Project Name\report.tsv" -l DIANN_file.tsv
```

By default, the script will create the output file in the same directory named “report_SQ_final_convert.tsv” unless the option -l is define in the command line.